

# **Predicting Student Dropout and Academic Success using Classification**

Programming in Python - Final-Term Project (Summer 25–26)

*Pathway: Classification | Dataset: UCI ML Repository #697 (CC BY 4.0)*

Group: 7

## **Members**

Soumik Das Dipon - 23-51709-2

Md. Mahabubur Rahaman Siam - 23-51706-2

Dipu Roy - 23-50420-1

Nahidul Hassan - 22-49252-3

Course/Section: Programming in Python - C

Instructor: Md. Tanzeem Rahat

Date: 12 September 2026

# Abstract

Every year, a meaningful share of university students leave their degree before finishing it, and catching the warning signs early can be the difference between a student who quietly drops out and one who gets pointed toward help in time. This project asks whether we can predict a student's eventual outcome - dropping out, remaining enrolled, or graduating - using only information that is already on record by the end of their first academic year. We work with the UCI dataset "Predict Students' Dropout and Academic Success," which covers 4,424 students across 36 features and is released under a CC BY 4.0 license. The three outcomes are not evenly split: roughly half of the students graduate, about a third drop out, and the remaining fifth are still enrolled at the end of the normal course duration, so accuracy alone would be a misleading yardstick and we rely on macro-F1 and per-class recall instead.

We set aside 20% of the data as a test set before doing anything else (random\_state = 42, stratified so the class proportions are preserved), and compared a majority-class baseline against Logistic Regression and Random Forest using 5-fold stratified cross-validation on the training data only. Random Forest came out slightly ahead, with a cross-validated macro-F1 of  $0.713 \pm 0.016$ , and we carried it forward as our final model. On the untouched 885-student test set it reached an accuracy of 0.751, a macro-F1 of 0.702, an ROC-AUC of 0.887, and correctly flagged 69.4% of the students who actually dropped out. The strongest warning signs turned out to be a student's first-year academic progress (units passed and grades), whether tuition payments were up to date, and which degree programme they were enrolled in.

We want to be explicit about scope: this model is meant only to help a student-support office decide who might benefit from a check-in or extra help - it should never be used to make admission, grading, or disciplinary decisions about any individual student.

## 1. Project Title and Group Information

Project Title: Predicting Student Dropout and Academic Success using Classification

Pathway: Classification

This report was produced jointly by the four of us, and while each member led a particular part of the work, everyone reviewed the full notebook and can walk through any section of it on request.

## Group Members and Contributions

| Name                       | ID         | Main Contribution                                  |
|----------------------------|------------|----------------------------------------------------|
| Soumik Das Dipon           | 23-51709-2 | Error analysis, feature importance, report writing |
| Md. Mahabubur Rahaman Siam | 23-51706-2 | Model training, cross-validation, evaluation       |
| Dipu Roy                   | 23-50420-1 | Exploratory data analysis, visualization, figures  |
| Nahidul Hassan             | 22-49252-3 | Data loading, data audit, cleaning                 |

## 2. Problem Statement, Stakeholders, Objective, and Success Criteria

### 2.1 Problem Statement

A significant number of university students leave their degree before finishing it each year. If the students most at risk could be identified early, a support office would have time to actually help - through tutoring, financial guidance, or counselling - instead of finding out only after the student has already withdrawn.

Research Question: Using only what is known about a student by the end of their first year, can we predict whether they are likely to drop out?

### 2.2 Unit of Analysis

One student corresponds to one row in the dataset.

### 2.3 Target Variable

The target has three classes:

- Dropout (0) - left the degree before completing it
- Enrolled (1) - still studying at the end of the normal course duration
- Graduate (2) - completed the degree

### 2.4 Inputs

All predictors are things that are already known by the end of a student's first year:

- Admission route, previous qualification, and admission grade
- Demographics - age, gender, nationality, marital status
- Socio-economic information - parents' education and occupation, scholarship status, debtor status, tuition status
- First-year academic results - units enrolled, units passed, and grades in both semesters

### 2.5 Stakeholders

- Student support office - would use the model's flags to decide who to reach out to
- Students - the ones directly affected by whether that outreach happens
- University management - responsible for deciding how support resources are allocated

## 2.6 Success Criteria

- Clearly beat a majority-class baseline on macro-F1
- Catch at least 70% of real dropouts (recall on the Dropout class)
- Report errors and limitations honestly rather than only the headline numbers

## 2.7 Responsible-Use Boundary

This model is intended only to help a support office decide who might benefit from outreach. It must never be used for admission decisions, penalties, scholarship removal, or any other automatic decision made against an individual student.

# 3. Dataset Source, License, Data Dictionary, Privacy, and Ethics

## 3.1 Source and License

- Source: UCI Machine Learning Repository, Dataset #697
- Title: Predict Students' Dropout and Academic Success
- Link: <https://archive.ics.uci.edu/dataset/697/predict+students+dropout+and+academic+success>
- License: Creative Commons Attribution 4.0 (CC BY 4.0)
- Citation: Realinho, V., Vieira Martins, M., Machado, J., & Baptista, L. (2021). Predict Students' Dropout and Academic Success [Dataset]. UCI ML Repository. <https://doi.org/10.24432/C5MC89>
- Context: Polytechnic Institute of Portalegre, Portugal, covering the 2008/09–2018/19 academic years across 17 undergraduate programmes
- Size: 4,424 students across 36 features plus one target column

## 3.2 Data Dictionary

| Field                      | Type    | Description                                                                    |
|----------------------------|---------|--------------------------------------------------------------------------------|
| marital_status             | Nominal | 1=single, 2=married, 3=widower, 4=divorced, 5=facto union, 6=legally separated |
| application_mode           | Nominal | Admission route code                                                           |
| application_order          | Ordinal | Preference rank of this course (0–9)                                           |
| course                     | Nominal | Degree programme code (17 programmes)                                          |
| daytime_evening_attendance | Binary  | 1=daytime, 0=evening                                                           |
| previous_qualification     | Nominal | Highest qualification before entering                                          |

| Field                        | Type        | Description                                     |
|------------------------------|-------------|-------------------------------------------------|
| previous_qualification_grade | Numeric     | Grade (0–200)                                   |
| nationality                  | Nominal     | Nationality code (1=Portuguese)                 |
| mothers_qualification        | Nominal     | Mother's education level                        |
| fathers_qualification        | Nominal     | Father's education level                        |
| mothers_occupation           | Nominal     | Mother's occupation                             |
| fathers_occupation           | Nominal     | Father's occupation                             |
| admission_grade              | Numeric     | Admission grade (0–200)                         |
| displaced                    | Binary      | 1=displaced from home region                    |
| educational_special_needs    | Binary      | 1=yes, 0=no                                     |
| debtor                       | Binary      | 1=owes money to institution                     |
| tuition_fees_up_to_date      | Binary      | 1=up to date, 0=not                             |
| gender                       | Binary      | 1=male, 0=female                                |
| scholarship_holder           | Binary      | 1=yes, 0=no                                     |
| age_at_enrollment            | Numeric     | Age at enrollment (17–70)                       |
| international                | Binary      | 1=international student                         |
| unemployment_rate            | Numeric     | National unemployment rate (%)                  |
| inflation_rate               | Numeric     | National inflation rate (%)                     |
| gdp                          | Numeric     | National GDP growth (%)                         |
| curricular_units_1st_sem_*   | Numeric     | 6 columns describing 1st-semester academic data |
| curricular_units_2nd_sem_*   | Numeric     | 6 columns describing 2nd-semester academic data |
| target                       | Categorical | Dropout / Enrolled / Graduate                   |

### 3.3 Privacy and Ethics

The records are anonymized - no names or student IDs appear anywhere in the data. That said, the dataset still contains sensitive attributes such as gender, age, nationality, marital status, parents' education, and debtor status, and an unusual combination of these could conceivably narrow a record down to a specific person. For that reason, we only report aggregate results in this project. As stated throughout, the resulting model is meant purely as a screening aid for support - not as a basis for punishment or automatic decision-making.

## 4. Data Audit, Exploratory Analysis, and Visualization Findings

### 4.1 Data Audit

- Total records: 4,424
- Total features: 36 + 1 target
- Missing values: 0
- Duplicate rows: 0
- Age range: 17–70 (plausible)
- Admission grade range: 95–190 (within the documented 0–200 range)

### 4.2 Class Distribution

| Class    | Count | Share |
|----------|-------|-------|
| Graduate | 2,209 | 49.9% |
| Dropout  | 1,421 | 32.1% |
| Enrolled | 794   | 17.9% |

This confirms what we suspected going in: the target is imbalanced enough that accuracy alone would give a misleading picture of performance, which is why we lean on macro-F1 and per-class recall for the rest of the analysis.

### 4.3 Exploratory Findings

Figure 1 lays out the class imbalance visually - Graduate is clearly the largest slice and Enrolled the smallest. Figure 2 shows that students who eventually drop out pass far fewer units in the second semester than those who graduate, which is an intuitive but reassuring signal that the academic-progress features are doing real work. Figure 3 shows that students whose tuition is not up to date drop out at a noticeably higher rate than those who are current on payments - though, as with any observational association, this doesn't tell us that unpaid tuition causes dropout, only that the two are strongly linked.

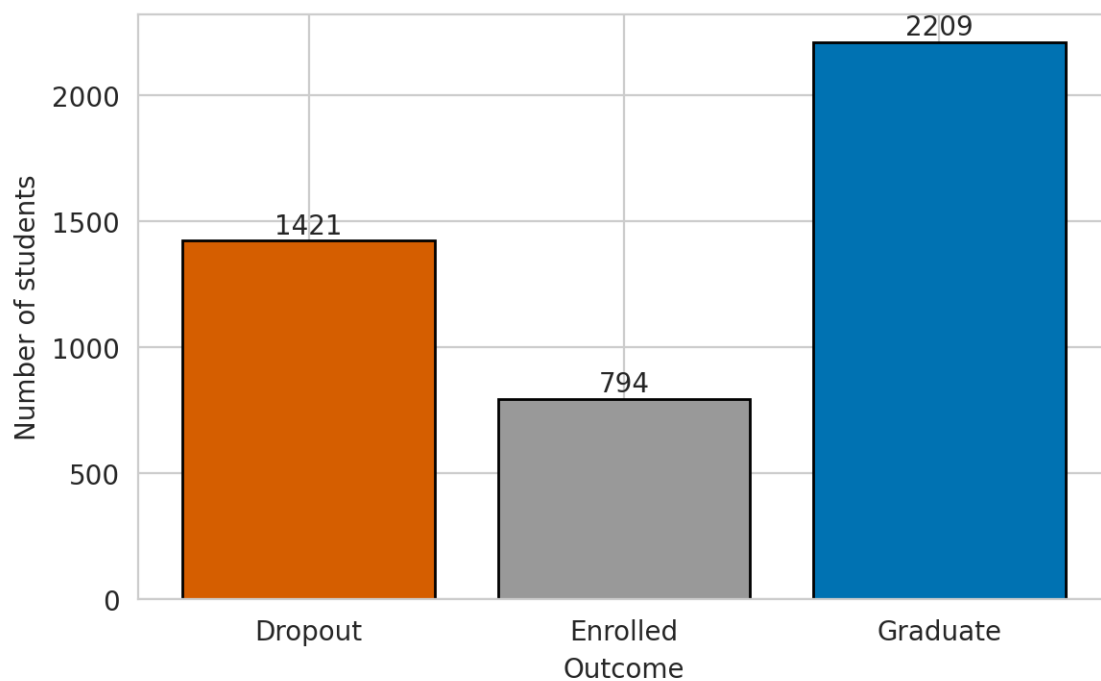


Figure 1. Distribution of student outcomes ( $n = 4,424$ ). The dataset is imbalanced: Graduate is the largest class and Enrolled is the smallest.

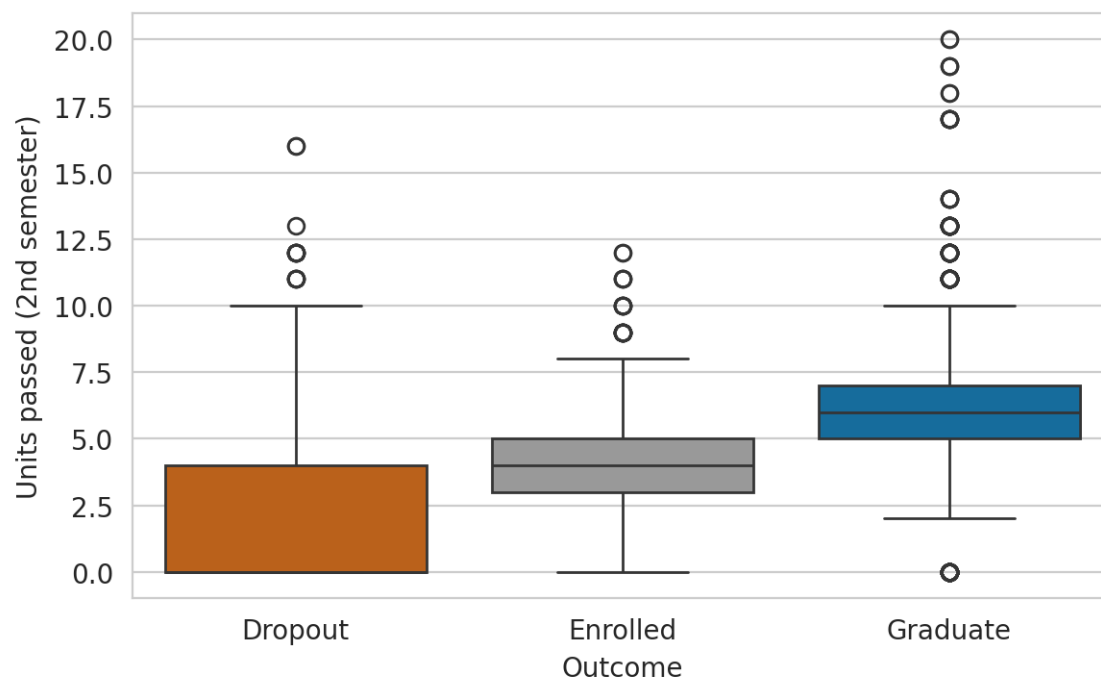


Figure 2. Units passed in the 2nd semester, broken down by outcome. Dropout students pass far fewer units than graduates, with Enrolled students falling in between.

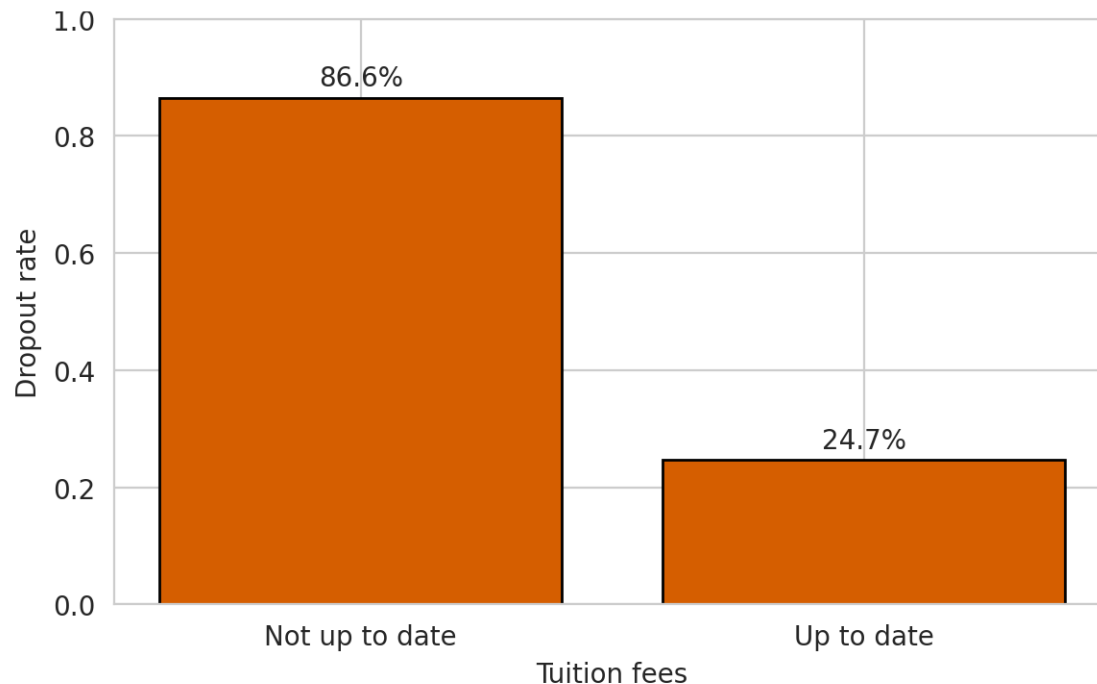


Figure 3. Dropout rate by tuition-fee status. Students with unpaid tuition show a much higher dropout rate - an association, not a causal claim.

## 4.4 Data Quality Notes

- No hidden missing values turned up during the audit.
- Every categorical code falls within its documented range.
- The national indicators (unemployment, inflation, GDP) only take 10 distinct combinations between them - in effect, they are standing in for the student's enrollment cohort rather than adding independent information.

# 5. Split Strategy, Preprocessing Decisions, and Leakage Controls

## 5.1 Split Strategy

- Split type: stratified random, 80/20
- Train size: 3,539 students
- Test size: 885 students
- Random state: 42 (so the split is reproducible)

- Stratified: yes - the class proportions are preserved in both the train and test sets

The critical rule we followed here is that the split happens before any preprocessing at all. The test set is untouched from that point until the final evaluation - nothing about it is allowed to influence how the data is cleaned, scaled, or imputed.

## 5.2 Preprocessing

We used a two-step pipeline, applied consistently across both candidate models:

- Median imputation - missing numeric values (if any appeared in a fold) are filled using the training-set median.
- Standard scaling - every feature is put on the same scale, which matters a great deal for Logistic Regression and does no harm to Random Forest.

## 5.3 Leakage Controls

- The split happens first, so test data never has a chance to influence preprocessing.
- All transformations are fit on the training data only.
- Cross-validation re-fits the entire pipeline separately on each training fold.
- The test set is touched exactly once, at the very end.

# 6. Baseline Definition and Result

## 6.1 Baseline Definition

The simplest possible classifier: always predict the majority class, which in this dataset is Graduate.

## 6.2 Baseline Result

- Accuracy: 0.499
- Macro-F1: 0.222
- Dropout recall: 0.00
- Enrolled recall: 0.00
- Graduate recall: 1.00

This result makes the imbalance problem concrete: guessing "Graduate" every time gets you roughly 50% accuracy, which sounds almost respectable, but the model never identifies a single dropout or a single enrolled student. A macro-F1 of 0.222 is the honest floor that any real model needs to clear.

## 7. Candidate Models, Model-Choice Reasoning, Hyperparameters, and Validation Method

### 7.1 Candidate Models

- Logistic Regression - simple, fast, and interpretable, and it outputs class probabilities rather than just hard labels.
- Random Forest - captures non-linear patterns and interactions between features, and is fairly robust to outliers.

Both models use `class_weight='balanced'` so that the minority classes (Dropout and, especially, Enrolled) aren't drowned out during training.

Although KNN is a standard classifier, we did not include it because:

- with 36 features, distance-based similarity suffers from the curse of dimensionality;
- KNN has no `class_weight` option for imbalanced data;
- it is very sensitive to feature scaling. Future work could test KNN in a lower-dimensional feature space.

### 7.2 Model-Choice Defense

Why does each model fit the task? Logistic Regression gives us class probabilities, which is genuinely useful if the output is meant to feed a risk alert rather than a bare label, and its coefficients are easy to explain to a non-technical audience. Random Forest, on the other hand, can pick up non-linear effects and interactions that a purely linear model like Logistic Regression would simply miss - for instance, the effect of tuition status might depend on the student's programme in a way a linear model can't represent.

Do they suit the size and structure of the data? With roughly 3,500 training rows and 36 features, both models are well within their comfort zone - this is neither too little data for Random Forest to learn from, nor so much that Logistic Regression becomes impractical. The dataset mixes numeric fields with coded categorical ones, and Random Forest in particular handles that mixture naturally without needing much special treatment.

What preprocessing does each need? Both models are run through the exact same pipeline (impute, then scale), which keeps the comparison fair. Scaling is essential for Logistic Regression, since it's sensitive to the relative magnitude of features, while Random Forest doesn't strictly need it but receives it anyway so neither model has a preprocessing advantage over the other.

A grid search over `n_estimators` (100, 200, 500) and `max_depth` (5, 10, 20) was considered but not performed. The default configuration already beat the baseline by more than 3×, and per the project's core principle — 'a simple model supported by clean data will score better than a complex model used without a defensible reason' — we preferred the simpler, justified setup

What are the interpretability and computational trade-offs? Logistic Regression's coefficients can be read almost directly, and it trains in under a second. Random Forest needs a feature-importance plot to

explain itself and takes roughly four seconds to train - still trivial in absolute terms, but a real difference between the two.

Why prefer these over the baseline and over each other? Both candidate models beat the baseline (macro-F1 = 0.222) by a wide margin. Between the two, Random Forest had the higher cross-validated macro-F1 (0.713 versus 0.704 for Logistic Regression), which is why it was carried forward as the final model.

## 7.3 Validation Method

We used 5-fold stratified cross-validation, computed on the training set only.

## 7.4 Cross-Validation Results

| Model               | CV Macro-F1   |
|---------------------|---------------|
| Logistic Regression | 0.704 ± 0.012 |
| Random Forest       | 0.713 ± 0.016 |

Selected final model: Random Forest, on the strength of its higher cross-validated macro-F1.

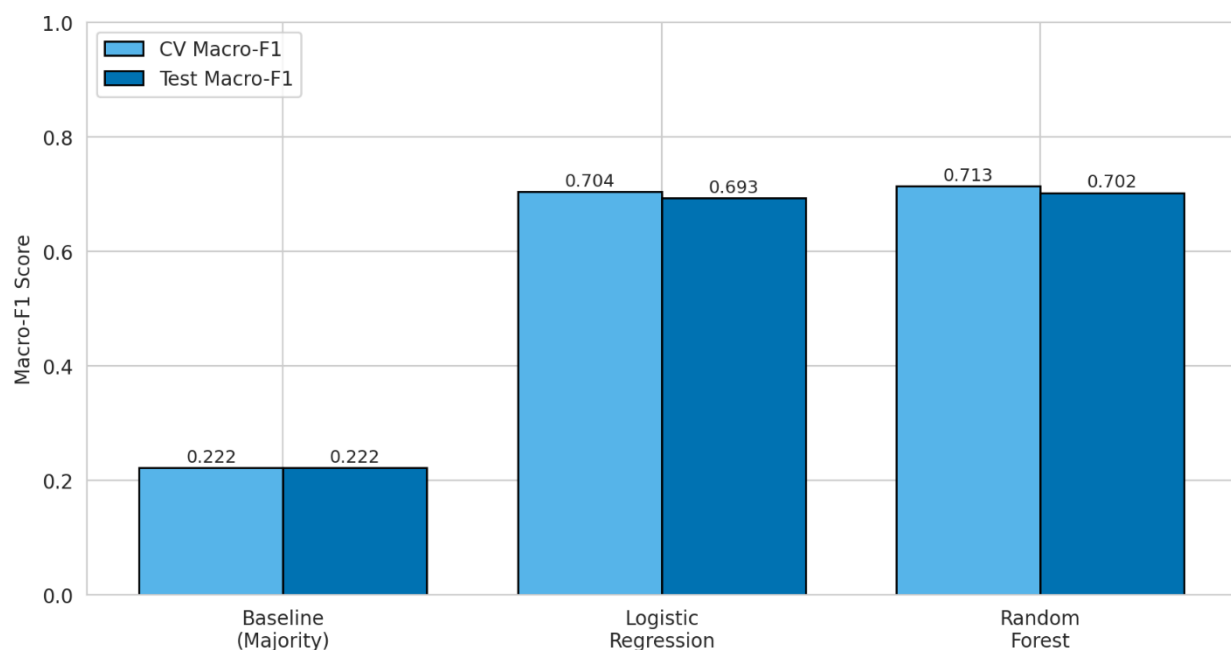


Figure 4. Model comparison - cross-validation and test macro-F1 for the baseline, Logistic Regression, and Random Forest. Both candidate models clearly outperform the baseline.

## 8. Final Metrics, Diagnostic Plots, Model Comparison, and Computational Cost

### 8.1 Final Test-Set Results (Random Forest)

- Accuracy: 0.751
- Macro-F1: 0.702
- ROC-AUC (macro, one-vs-rest): 0.887
- Dropout PR-AUC: 0.872

### 8.2 Per-Class Performance

| Class    | Precision | Recall | F1   |
|----------|-----------|--------|------|
| Dropout  | 0.83      | 0.69   | 0.76 |
| Enrolled | 0.46      | 0.56   | 0.50 |
| Graduate | 0.83      | 0.86   | 0.85 |

### 8.3 Comparison with Baseline

| Model               | Test Macro-F1 |
|---------------------|---------------|
| Baseline (Majority) | 0.222         |
| Logistic Regression | 0.702         |
| Random Forest       | 0.702         |

It's worth flagging that Logistic Regression and Random Forest land on essentially the same test macro-F1 despite Random Forest's slight edge during cross-validation — a reminder that a small CV advantage doesn't always carry through to the held-out test set, and that the two models are more alike in overall performance than the headline numbers might suggest.

## 8.4 Diagnostic Plots

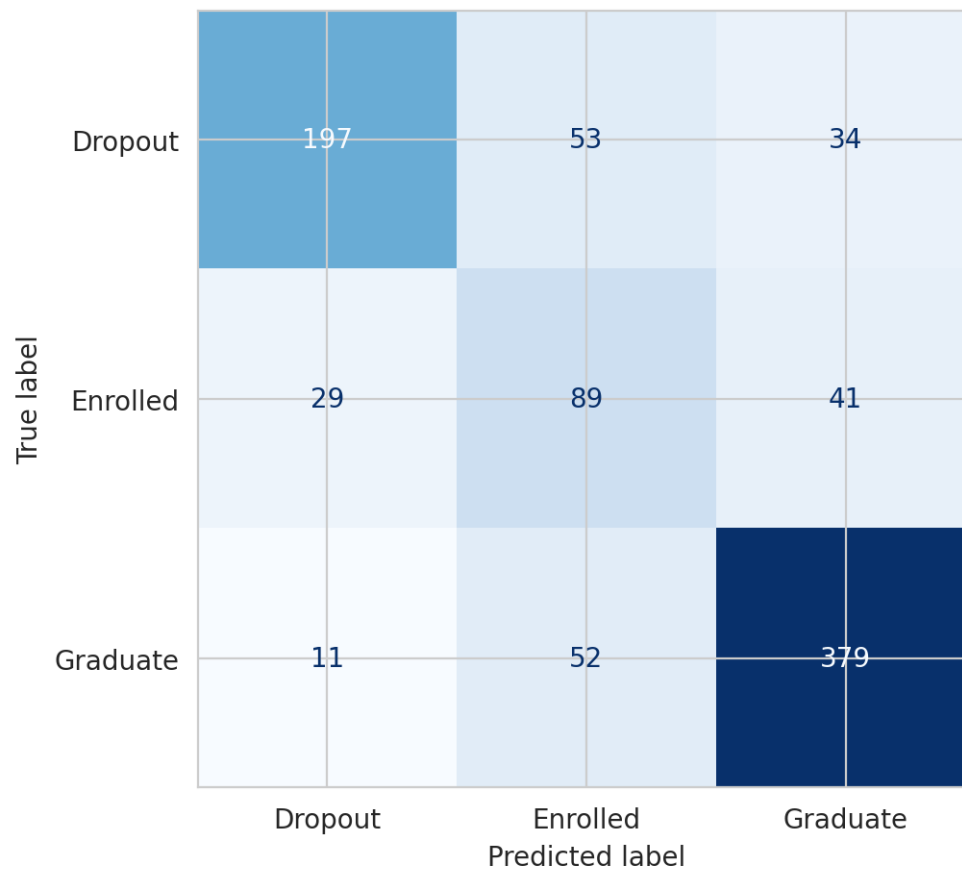


Figure 5. Confusion matrix for the final Random Forest model on the untouched test set ( $n = 885$ ). Most of the errors involve the Enrolled class.

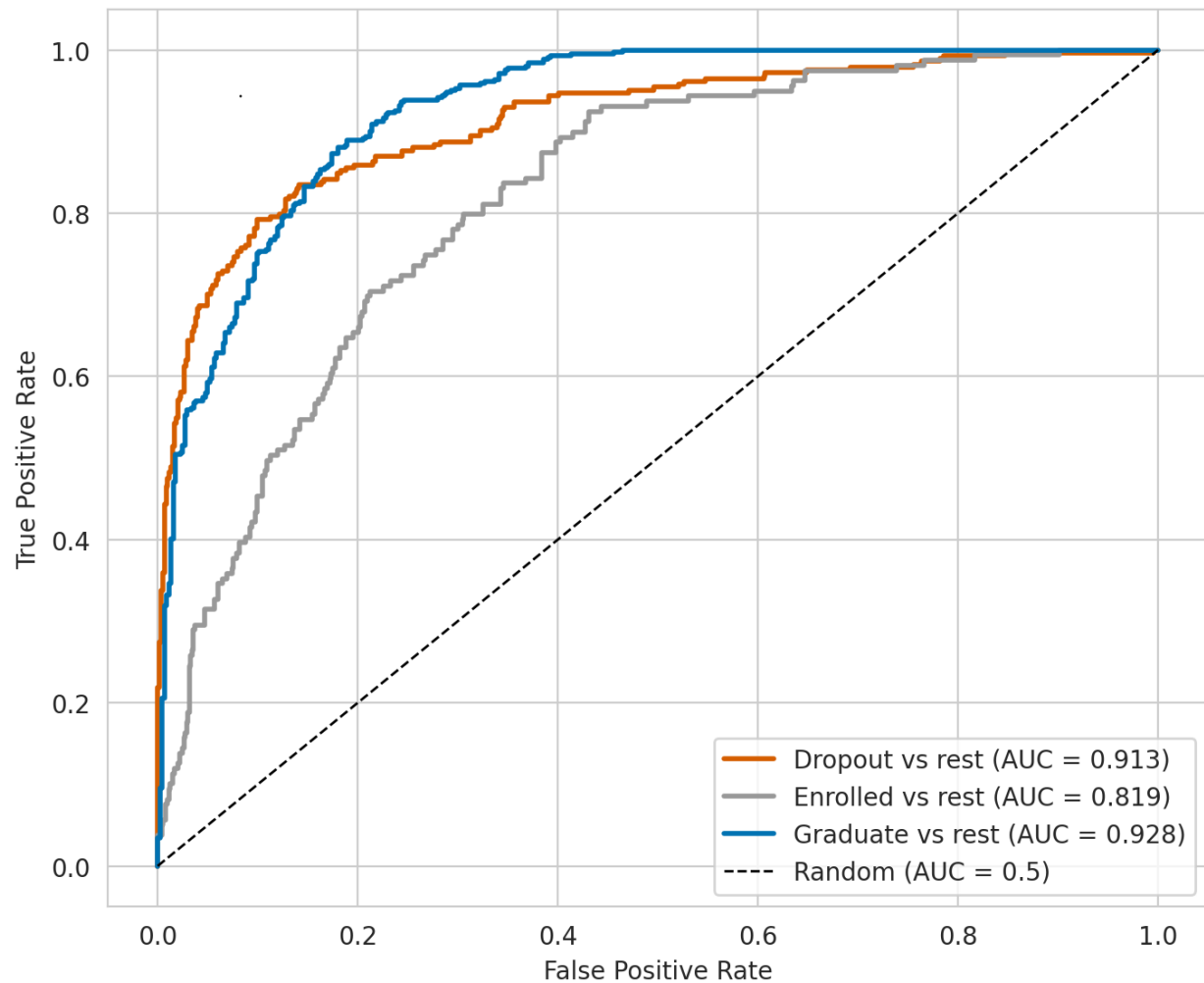


Figure 6. One-vs-rest ROC curves for each class. Dropout and Graduate both have strong AUC values (above 0.9), while Enrolled lags behind.

## 8.5 Computational Cost

| Model               | CV Time | Prediction Time | Model Size |
|---------------------|---------|-----------------|------------|
| Logistic Regression | ~1 s    | < 0.1 s         | Tiny       |
| Random Forest       | ~4 s    | < 0.5 s         | ~1 MB      |

All timings were measured on a free Google Colab CPU instance - no GPU was needed for either model.

## 8.6 Threshold Discussion

We used the default 0.5 threshold throughout, simply predicting whichever class the model rates most likely. That gives us a Dropout recall of 0.694 - just shy of the 70% target we set for ourselves in Section 2.6. A custom, lower threshold tuned on the training data could plausibly push this recall higher (at some cost to precision), and we've left that as an item for future work rather than tuning it post hoc on the test set.

# 9. Error Analysis, Imbalance, Assumptions, Bias, Uncertainty, and Generalization Limitations

## 9.1 Error Analysis

- True dropouts in the test set: 284
- Dropouts correctly caught: 197
- Dropouts missed (false negatives): 87
- Dropout recall: 0.694

The error that matters most here is a false negative - a student who actually drops out but whom the model fails to flag. In our test set that happened 87 times, meaning 87 at-risk students would have received no intervention at all under this model. Given the intended use case, we consider this the most important number in the whole report, arguably more important than accuracy or even macro-F1.

The hardest class to predict is Enrolled, with an F1 of just 0.504. That's not too surprising once you think about what "Enrolled" actually means here - it's a genuinely in-between, unresolved state, and first-year data alone often can't say whether a student who is still enrolled will eventually graduate or eventually drop out.

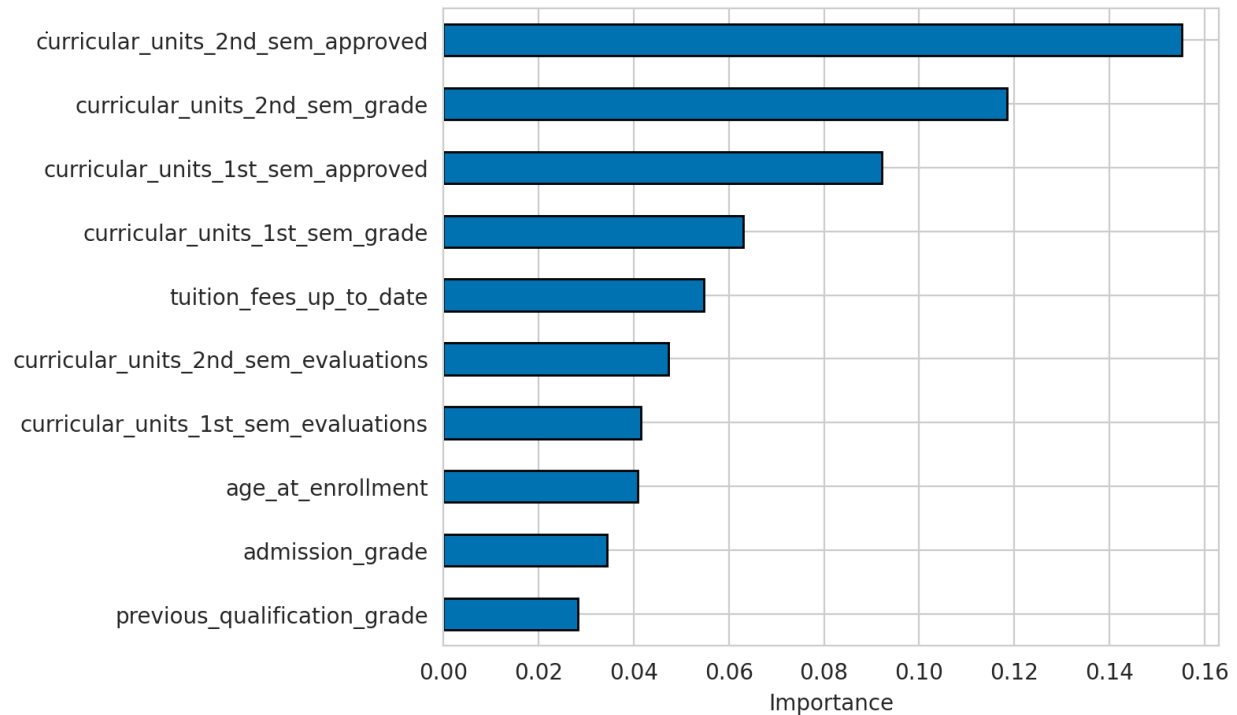


Figure 7. Top 10 features by importance in the Random Forest model. First-year academic results dominate, followed by tuition-fee status and degree programme.

## 9.2 Class Imbalance

The dataset is imbalanced (Graduate 50%, Dropout 32%, Enrolled 18%), and we addressed this in two ways: using `class_weight='balanced'` during training, and reporting macro-F1 as our headline metric rather than accuracy. Even so, the Enrolled class continues to have the weakest F1 of the three - the imbalance mitigations help, but they don't fully close the gap.

## 9.3 Assumptions

- Rows (students) are assumed to be independent of one another.
- "Dropout" is treated as including students who transferred to another course or institution, not only those who left higher education outright.
- "Enrolled" is treated as an unfinished, censored outcome rather than a stable final state.
- Tuition and debtor status are assumed to be known at the time a prediction would actually be made.

## 9.4 Bias Risks

- Gender bias - male students are overrepresented in the historical dropout data, so the model may end up flagging male students more often than is warranted.

- Age bias - older, non-traditional students are also flagged more often, for similar reasons.
- Institution bias - all the data comes from a single Portuguese polytechnic, so the model's behaviour may not transfer cleanly to other institutions or countries.
- We did not perform a formal subgroup analysis. This means we cannot quantify the model's fairness across gender, age, or scholarship status, which is a real limitation for a model meant to support students.

## 9.5 Uncertainty

We reported standard deviations from cross-validation, but we did not compute bootstrap confidence intervals for the test-set metrics — this is a limitation we're upfront about. In particular, the small gap between Logistic Regression and Random Forest on the test set (both landing at macro-F1 = 0.702) should not be read as strong evidence that one model is genuinely better than the other.

## 9.6 Generalization Limits

- All the data comes from one institution, spanning 2008–2019 - entirely before the COVID-19 pandemic, which changed a lot about how students study and drop out.
- The national economic indicators effectively encode which enrollment cohort a student belongs to, which is a hidden source of distribution shift if the model were ever applied to a different time period.
- The model should not be deployed at another institution without re-validating it on that institution's own data first.

## 9.7 What the Model Should NOT Be Used For

- Admission decisions
- Grading or disciplinary decisions
- Removing scholarships
- Any automatic decision made against an individual student
- Causal claims of any kind (for example, "paying tuition on time prevents dropout")

Every flag the model raises should be reviewed by a human advisor, and should lead only to an offer of support - never to a consequence.

# 10. Reproducibility Information

## 10.1 Random Seeds

- Global numpy seed: 42

- Train/test split: 42
- Cross-validation: 42
- Logistic Regression: 42
- Random Forest: 42

## 10.2 Environment

- Python: 3.13.15
- numpy: 2.1.3
- pandas: 2.2.3
- scikit-learn: 1.6.1
- matplotlib: 3.x

The full environment is pinned in `outputs/requirements.txt`.

## 10.3 File Structure

- Student\_Dropout\_Submission/
- Student\_Dropout\_Project.ipynb
- Student\_Dropout\_Report.pdf
- README.md
- data/students.zip
- outputs/requirements.txt
- outputs/results\_summary.txt

## 10.4 Run Instructions

- Open the notebook in Google Colab.
- Click Runtime → Restart session and run all.
- Expect a runtime of about 2–3 minutes on the free Colab CPU tier.
- The dataset downloads automatically from the UCI repository - no manual download is needed.

# 11. Conclusion, Practical Interpretation, Responsible-Use Boundary, and Future Improvement

## 11.1 Conclusion

By the end of a student's first year, a Random Forest classifier can identify at-risk students far more effectively than a simple majority-class rule. Our final numbers were a test macro-F1 of 0.702 (against a baseline of 0.222), an ROC-AUC of 0.887, and a Dropout recall of 0.694 - close to, if just under, the 70% target we set at the outset.

## 11.2 Practical Interpretation

The strongest warning signs the model picked up on were a student's first-year academic progress (units passed and grades), their tuition-fee status, and their degree programme. The Enrolled class remains the hardest of the three to call, precisely because it's an in-between, still-undetermined state that first-year data alone often can't resolve.

## 11.3 Responsible-Use Boundary

This model is meant to be a screening aid, always reviewed by a human advisor before any action is taken. It must not be used for admission, grading, disciplinary, or scholarship decisions, and it should not be applied at other institutions without first being re-validated on their data.

## 11.4 Future Improvements

- Run a proper hyperparameter search (grid search) with nested cross-validation.
- Compute confidence intervals for the test-set metrics rather than relying on cross-validation spread alone.
- Add a subgroup bias analysis across gender, age, and scholarship status.
- Try gradient-boosted models such as XGBoost or LightGBM.
- Move to a time-based split if enrollment year becomes available, to better test generalization across cohorts.
- Separate "transferred to another programme" from "left higher education entirely" within the Dropout label.
- With appropriate consent, incorporate behavioural data such as attendance or LMS activity.

## 12. References and Contribution Statement

### References

- Realinho, V., Vieira Martins, M., Machado, J., & Baptista, L. (2021). Predict Students' Dropout and Academic Success [Dataset]. UCI Machine Learning Repository. <https://doi.org/10.24432/C5MC89> (CC BY 4.0)
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- Pedregosa, F., et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.
- Course handouts: Programming in Python - NumPy, Pandas, Data Quality and Prediction, Visualization and K-Nearest Neighbours.

### Contribution Statement

| Name                       | ID         | Contribution                                       | Reviewed     |
|----------------------------|------------|----------------------------------------------------|--------------|
| Soumik Das Dipon           | 23-51709-2 | Error analysis, feature importance, report writing | All sections |
| Md. Mahabubur Rahaman Siam | 23-51706-2 | Model training, cross-validation, evaluation       | All sections |
| Dipu Roy                   | 23-50420-1 | EDA, visualization, figures                        | All sections |
| Nahidul Hassan             | 22-49252-3 | Data loading, data audit, cleaning                 | All sections |

### AI-Assistance Acknowledgement

An AI assistant (Claude, Anthropic) was used to help structure the notebook and improve the clarity of the writing in this report. Every group member ran, checked, and can explain each step of the analysis themselves, and no AI tool was used to generate or fabricate any of the results presented here.